CHAPTER 8

SMALL SAMPLE TESTS

CHAPTER ESSENCE AT A GLANCE:
Master exact sampling tests for small samples (n < 30): Student's t-test for single mean, difference of two independent means, paired t-test for dependent samples, and Snedecor's F-test.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Student t-Distribution and Degrees of Freedom:** Discovered by W.S. Gosset ('Student'); symmetric bell-shaped curve with heavier tails than normal; degrees of freedom (df = n - 1) determine exact shape.

• **One-Sample and Two-Sample t-Test:** One-sample t = (x_{bar} - mu0) divided by [s divided by sqrt(n)] with df = n - 1; Two independent samples pooled t = (x_{bar1} - x_{bar2}) divided by [s_p * sqrt(1 divided by n1 + 1 divided by n2)] with df = n1 + n2 - 2.

• **Paired t-Test for Dependent Observations:** Applied to 'before-and-after' studies on the same subjects; examines differences d = x - y; t = d_{bar} divided by [s_d divided by sqrt(n)] with df = n - 1.

• **Snedecor F-Test for Equality of Variances:** Tests H0: $sigma1^2$ = $sigma2^2$; F = $s1^2$ divided by $s2^2$ (placing larger variance in numerator so F >= 1) with df = (n1 - 1, n2 - 1).

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \sim t_{(n-1)} \quad \text{where} \quad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

Student One-Sample t-Test Statistic

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} \rightarrow t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t_{(n_1+n_2-2)}$$

Two-Sample Pooled Variance t-Test Formulation

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}} \sim t_{(n-1)} \quad \text{where} \quad d_i = x_i - y_i$$

Paired Sample Difference t-Test for Dependent Pairs

$$F = \frac{s_1^2}{s_2^2} \sim F_{(n_1-1,\ n_2-1)} \quad [s_1^2 \geq s_2^2]$$

Snedecor Variance Ratio Test Statistic

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In independent two-sample t-tests, degrees of freedom is n1 + n2 - 2; in paired t-tests, degrees of freedom is n - 1 (where n is the number of pairs).

• **Alert:** In the F-test, always place the larger sample variance in the numerator so that calculated F is greater than or equal to 1.

• **Alert:** As degrees of freedom tend to infinity, the Student t-distribution converges to the Standard Normal Z-distribution.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State two basic assumptions required for applying Student's t-test. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Assumption 1:** The parent population from which the sample is drawn is normally distributed. **[1 Mark]**

• **Assumption 2:** Sample observations are random and mutually independent, and sample size is small (n < 30) with unknown population variance. **[1 Mark]**

2. A sample of 10 measurements of the diameter of a sphere gave a mean of 4.38 cm and a sample standard deviation s = 0.06 cm. Test the hypothesis that the true diameter is 4.35 cm at 5 percent level of significance (given t for 9 df at 5 percent is 2.262). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Hypotheses:** H0: $\mu = 4.35$ cm. H1: $\mu \neq 4.35$ cm (Two-tailed test). df = n - 1 = 10 - 1 = 9. Critical value $t_{0}.05 = 2.262$. **[1 Mark]**

• **Test Statistic:** t = (x_{bar} - μ) divided by [s divided by square root of n] = (4.38 - 4.35) divided by [0.06 divided by square root of 10]. **[1 Mark]**

• **Calculation:** square root of 10 approx 3.162. Denominator = 0.06 divided by 3.162 = 0.01897. t = 0.03 divided by 0.01897 = 1.58. **[1 Mark]**

• **Conclusion:** Since calculated |t| = 1.58 < 2.262, it falls in the acceptance region. We fail to reject H0 at 5 percent level. The true diameter can be accepted as 4.35 cm. **[1 Mark]**

3. To test the effect of a training program on sales, the sales of 5 salesmen before and after training were recorded: Salesman A (Before=20, After=24), B (18, 22), C (22, 26), D (20, 22), E (25, 26). Test whether the training program was effective at 5 percent level (given t for 4 df at 5 percent one-tailed is 2.132). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Hypotheses:** H0: $mu_d = 0$ (training has no effect). H1: $mu_d > 0$ (training increases sales, one-tailed test). **[1 Mark]**

• **Differences (After - Before):** d = 4, 4, 4, 2, 1. Sum of d = 15. Mean difference $d_{bar} = 15$ divided by 5 = 3. **[1 Mark]**

• **Variance of d:** (d - d_{bar}) = +1, +1, +1, -1, -2. Squared = 1, 1, 1, 1, 4. Sum = 8. $s_d^2 = 8$ divided by (5 - 1) = 8 divided by 4 = 2. s_d = square root of 2 = 1.414. **[1 Mark]**

• **Calculated t:** t = d_{bar} divided by [s_d divided by square root of n] = 3 divided by [1.414 divided by square root of 5] = 3 divided by [1.414 divided by 2.236] = 3 divided by 0.632 = 4.74. Since calculated t = 4.74 > 2.132, reject H0. The training program is significantly effective. **[1 Mark]**

4. Two independent samples of sizes 9 and 11 had variances 36 and 16 respectively. Test whether the two populations have equal variances at 5 percent level of significance (given critical F for df (8, 10) is 3.07). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Hypotheses:** H0: $sigma1^2 = sigma2^2$. H1: $sigma1^2 \neq sigma2^2$. df1 = 9 - 1 = 8, df2 = 11 - 1 = 10. **[1 Mark]**

• **Calculated F:** F = $s1^2$ divided by $s2^2$ = 36 divided by 16 = 2.25 (larger variance in numerator). **[1 Mark]**

• **Conclusion:** Since calculated F = 2.25 < critical F = 3.07, we fail to reject H0. The two population variances can be regarded as equal. **[1 Mark]**

5. What are the degrees of freedom for: (a) A two-sample pooled t-test with sample sizes n1 = 12 and n2 = 15? (b) A paired t-test with 10 pairs of observations? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) Two-Sample Pooled t-Test:** df = n1 + n2 - 2 = 12 + 15 - 2 = 25. **[1 Mark]**

• **(b) Paired t-Test:** df = n - 1 = 10 - 1 = 9. **[1 Mark]**

6. When is a paired t-test applied instead of an independent two-sample t-test? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Application Criterion:** Applied when sample observations are naturally paired or dependent, typically measuring the same experimental units before and after a treatment to eliminate individual subject variability. **[2 Marks]**